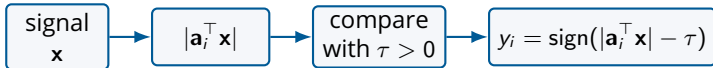


Phase Retrieval under One-Bit Quantization

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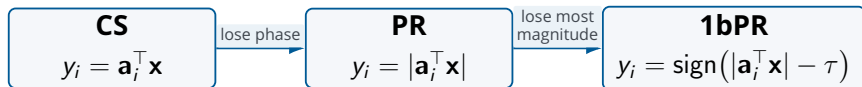
SIAM Annual Meeting
Recent Advances in Computational Data Science
Cleveland, Ohio
July 6, 2026
Joint work with Ming Yuan



Introduction

Measurements: CS, PR, and 1bPR

$$\mathbf{x} \in \mathbb{R}^n, \quad \mathbf{a}_1, \dots, \mathbf{a}_m \sim N(0, \mathbf{I}_n)$$



Phase retrieval

$$y_i = |\mathbf{a}_i^\top \mathbf{x}|$$

Recover $\mathbf{x}$ up to the unavoidable ambiguity $\mathbf{x} \sim -\mathbf{x}$.

One-bit phase retrieval

$$y_i = \text{sign}(|\mathbf{a}_i^\top \mathbf{x}| - \tau)$$

Keep only whether the magnitude crosses a threshold.

CS: exact recovery

PR: recovery of $\pm \mathbf{x}$

1bPR: still recoverable

Main Questions

$$0 < \alpha \leq \|\mathbf{x}\|_2 \leq \beta < \infty, \quad \text{dist}(\widehat{\mathbf{x}}, \mathbf{x}) := \min\{\|\widehat{\mathbf{x}} - \mathbf{x}\|_2, \|\widehat{\mathbf{x}} + \mathbf{x}\|_2\}.$$

For concreteness: $\alpha = \frac{1}{2}, \quad \beta = 2, \quad \tau = 1.$

Question 1: Statistical / information-theoretic

What is the best possible recovery error rate from $\mathbf{y} = \text{sign}(|\mathbf{Ax}| - \tau)$?

$$\inf_{\widehat{\mathbf{x}}} \sup_{\alpha \leq \|\mathbf{x}\|_2 \leq \beta} \text{dist}(\widehat{\mathbf{x}}, \mathbf{x}) = ?$$

Question 2: Algorithmic / computational

Can an efficient algorithm achieve the optimal statistical rate?

optimal rate + efficient computation ?

Question 3: Sparse Vectors?

Main focus: unstructured signals.

Results extend to sparse signals.

One-bit phase retrieval — $\mathbf{y} = \text{sign}(|\mathbf{Ax}| - \tau)$

- ▶ Empirical algorithms, but without non-asymptotic guarantees: (Eamraz et al., 2022; Kishore and Seelamantula, 2020; Mukherjee and Seelamantula, 2018).
- ▶ Existing theory gives a suboptimal rate:

$$\text{dist}(\hat{\mathbf{x}}_{\text{spectral}}, \mathbf{x}) = \tilde{O}\left(\sqrt{\frac{n}{m}}\right) \quad (\text{Domel-White and Bodmann, 2022}).$$

- ▶ Related low-dose phase retrieval setting: (Römer and Krahmer, 2024).

One-bit compressed sensing — $\mathbf{y} = \text{sign}(\mathbf{Ax})$, $\mathbf{x} \in \Sigma_k^n$

- ▶ Statistical optimal rate:

$$\tilde{\Theta}\left(\frac{k}{m}\right) \quad (\text{Jacques et al., 2013}).$$

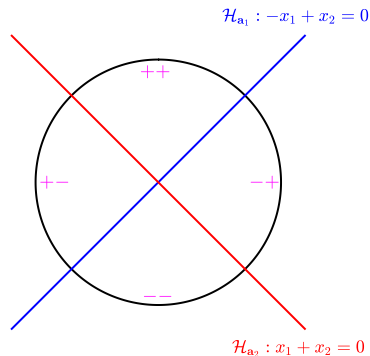
- ▶ Efficient algorithmic recovery via NBIHT: (Matsumoto and Mazumdar, 2024).

Outline

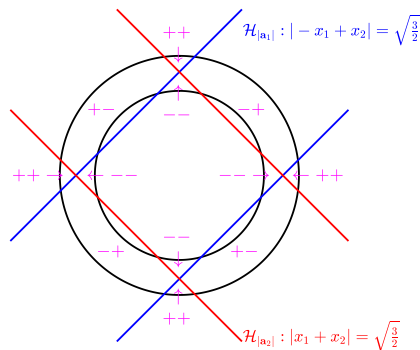
- 1 Introduction
- 2 Statistical Result
- 3 Computational Result
- 4 Technical Overview
- 5 Experiments
- 6 Sparse Case
- 7 Conclusion

Statistical Result

Hyperplane Tessellation



1bCS



1bPR

$$\mathcal{H}_{\mathbf{a}} = \{\mathbf{u} \in \mathbb{R}^n : \mathbf{a}^\top \mathbf{u} = 0\}$$

$$\mathcal{H}_{|\mathbf{a}|} = \{\mathbf{u} \in \mathbb{R}^n : |\mathbf{a}^\top \mathbf{u}| = \tau\}$$

Each y_i asks which side of a (phaseless) hyperplane contains $\mathbf{x}$.

Hamming Distance (Loss) Minimization

Optimal decoder — hamming distance (loss) minimization:

$$\begin{aligned}\hat{\mathbf{x}}_{\text{HDM}} &\in \arg \min_{\alpha \leq \|\mathbf{u}\|_2 \leq \beta} d_H(\text{sign}(|\mathbf{A}\mathbf{u}| - \tau), \mathbf{y}) \\ &= \arg \min_{\alpha \leq \|\mathbf{u}\|_2 \leq \beta} \sum_{i=1}^m \mathbf{1} \left(\text{sign}(|\mathbf{a}_i^T \mathbf{u}| - \tau) \neq \text{sign}(|\mathbf{a}_i^T \mathbf{x}| - \tau) \right) \\ &= \left\{ \alpha \leq \|\mathbf{u}\|_2 \leq \beta : \text{sign}(|\mathbf{A}\mathbf{u}| - \tau) = \text{sign}(|\mathbf{A}\mathbf{x}| - \tau) \right\}\end{aligned}$$

$$\text{sign}(|\mathbf{A}\mathbf{x}| - \tau) \quad \begin{array}{|c|c|c|c|c|c|c|c|} \hline + & - & + & + & - & + & - & + \\ \hline \end{array}$$

$$\text{sign}(|\mathbf{A}\mathbf{u}| - \tau) \quad \begin{array}{|c|c|c|c|c|c|c|c|} \hline + & - & - & + & - & + & + & + \\ \hline \end{array}$$

$$\text{sign}(|\mathbf{A}\hat{\mathbf{x}}_{\text{HDM}}| - \tau) = \text{sign}(|\mathbf{A}\mathbf{x}| - \tau)$$

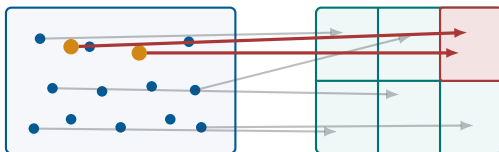
Optimal Error Rate

(HDM) Upper bound - binary embedding (Dirksen and Mendelson, 2021; Oymak and Recht, 2015)

$$\sup_{\alpha \leq \mathbf{x} \leq \beta} \text{dist}(\widehat{\mathbf{x}}, \mathbf{x}) \lesssim_{\alpha, \beta, \tau} \frac{n}{m} \log \frac{m}{n}$$

(Counting) Lower bound - pigeonhole principle

$$\sup_{\alpha \leq \mathbf{x} \leq \beta} \text{dist}(\widehat{\mathbf{x}}, \mathbf{x}) \gtrsim_{\alpha, \beta, \tau} \frac{n}{m}$$



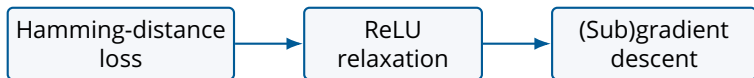
Up to log factor, the optimal (uniform) error rate is $\frac{n}{m}$

Computational Result

From HDM to (Sub)Gradient Descent

$$\hat{\mathbf{x}}_{\text{HDM}} \in \arg \min_{\alpha \leq \|\mathbf{u}\|_2 \leq \beta} \mathcal{L}_{\text{hd}}(\mathbf{u}), \quad \mathcal{L}_{\text{hd}}(\mathbf{u}) := \frac{1}{m} \sum_{i=1}^m \mathbf{1}(-y_i(|\mathbf{a}_i^\top \mathbf{u}| - \tau) > 0)$$

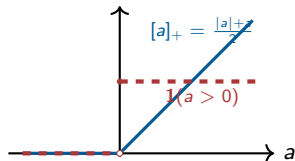
Computational challenges: nonconvex annulus constraint +
discontinuous 0-1 (Hamming) loss



$$s_i(\mathbf{u}) := -y_i(|\mathbf{a}_i^\top \mathbf{u}| - \tau)$$

$$\mathcal{L}_{\text{hd}}(\mathbf{u}) = \frac{1}{m} \sum_{i=1}^m \mathbf{1}(s_i(\mathbf{u}) > 0)$$

$$\mathcal{L}_{\text{ReLU}}(\mathbf{u}) = \frac{1}{2m} \sum_{i=1}^m (|s_i(\mathbf{u})| + s_i(\mathbf{u}))$$



(Sub)Gradient Descent with ReLU Loss

$$\mathcal{L}_{\text{ReLU}}(\mathbf{u}) = \frac{1}{2m} \sum_{i=1}^m \left(\left| \mathbf{a}_i^\top \mathbf{u} - \tau \right| - y_i (\mathbf{a}_i^\top \mathbf{u} - \tau) \right)$$

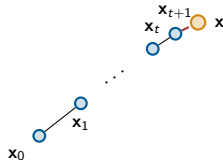
still **nonconvex** but **nearly differentiable**

$$\partial \mathcal{L}_{\text{ReLU}}(\mathbf{u}) = \frac{1}{2m} \sum_{i=1}^m \left(\text{sign}(\mathbf{a}_i^\top \mathbf{u} - \tau) - \text{sign}(\mathbf{a}_i^\top \mathbf{x} - \tau) \right) \text{sign}(\mathbf{a}_i^\top \mathbf{u}) \mathbf{a}_i := \mathbf{h}(\mathbf{u}, \mathbf{x})$$

Subgradient descent

Given an initial point $\mathbf{x}_0$ and a step size $\eta > 0$,

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \mathbf{h}(\mathbf{x}_t, \mathbf{x}).$$



Algorithmic Result

Convergence of GD (Chen and Yuan, 2026, Theorem 2.4)

Fix (α, β, τ) . If

- 1 $\text{dist}(\mathbf{x}_0, \mathbf{x}) \lesssim \frac{1}{\sqrt{\log(m/n)}};$ (**spectral method**)
- 2 $\eta = \sqrt{\frac{\pi e}{2}}\tau;$ (**good η under Gaussian A**)
- 3 $m \gtrsim n;$ (**optimal**)

Then w.h.p., for all $\mathbf{x} \in \mathbb{A}_\alpha^\beta = \{\mathbf{u} : \alpha \leq \|\mathbf{u}\|_2 \leq \beta\},$

$$\text{dist}(\mathbf{x}^{(t)}, \mathbf{x}) \leq (1 - c_1)^t + O\left(\frac{n}{m} \log^2\left(\frac{m}{n}\right)\right)$$

Subgradient descent (with spectral initialization) reaches the near-optimal n/m up to logarithmic factors.

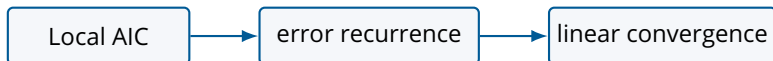
Technical Overview

Approximate Invertibility Condition (AIC)

Local AIC - a Sufficient Condition

For all $\mathbf{u}, \mathbf{v} \in \mathbb{A}_{\alpha, \beta}$ with $\|\mathbf{u} - \mathbf{v}\|_2 \lesssim 1/\sqrt{\log(m/n)}$,

$$\|\mathbf{u} - \mathbf{v} - \eta \mathbf{h}(\mathbf{u}, \mathbf{v})\|_2 \leq c_1 \|\mathbf{u} - \mathbf{v}\|_2 + \sqrt{\tilde{O}\left(\frac{n}{m}\right) \|\mathbf{u} - \mathbf{v}\|_2} + \tilde{O}\left(\frac{n}{m}\right).$$



Taking $\mathbf{u} = \mathbf{x}_t$ and $\mathbf{v} = \mathbf{x}$, we have

$$\mathbf{u} - \eta \mathbf{h}(\mathbf{u}, \mathbf{v}) = \mathbf{x}_t - \eta \mathbf{h}(\mathbf{x}_t, \mathbf{x}) = \mathbf{x}_{t+1}.$$

Hence Local AIC gives the one-step error recursion

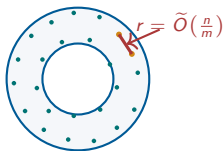
$$\|\mathbf{x}_{t+1} - \mathbf{x}\|_2 \leq c_1 \|\mathbf{x}_t - \mathbf{x}\|_2 + \sqrt{\tilde{O}\left(\frac{n}{m}\right) \|\mathbf{x}_t - \mathbf{x}\|_2} + \tilde{O}\left(\frac{n}{m}\right).$$

Analyzing this scalar recurrence yields linear convergence down to the optimal error rate $\tilde{O}(\frac{n}{m})$.

Discretization

$$\|\mathbf{u} - \mathbf{v} - \eta \mathbf{h}(\mathbf{u}, \mathbf{v})\|_2 \leq c_1 \|\mathbf{u} - \mathbf{v}\|_2 + \sqrt{\tilde{O}\left(\frac{n}{m}\right) \|\mathbf{u} - \mathbf{v}\|_2} + \tilde{O}\left(\frac{n}{m}\right),$$

$$\forall (\mathbf{u}, \mathbf{v}) \in \mathbb{A}_{\alpha, \beta} \times \mathbb{A}_{\alpha, \beta}, \quad \|\mathbf{u} - \mathbf{v}\|_2 \lesssim \frac{1}{\sqrt{\log(m/n)}}.$$



$$r = \tilde{O}\left(\frac{n}{m}\right), \quad \mathcal{N}_r \subset \mathbb{A}_{\alpha, \beta} \text{ is an } r\text{-net.}$$

A discretization argument reduces the proof from $\mathbb{A}_{\alpha, \beta} \times \mathbb{A}_{\alpha, \beta}$ to $\mathcal{N}_r \times \mathcal{N}_r$

$$\|\mathbf{u} - \mathbf{v} - \eta \mathbf{h}(\mathbf{u}, \mathbf{v})\|_2 \leq c_1 \|\mathbf{u} - \mathbf{v}\|_2 + \sqrt{\tilde{O}\left(\frac{n}{m}\right) \|\mathbf{u} - \mathbf{v}\|_2} + \tilde{O}\left(\frac{n}{m}\right),$$

$$\forall (\mathbf{u}, \mathbf{v}) \in \mathcal{N}_r \times \mathcal{N}_r, \quad \|\mathbf{u} - \mathbf{v}\|_2 \lesssim \frac{1}{\sqrt{\log(m/n)}}.$$

Decompose the Subgradient

$$\mathbf{h}(\mathbf{u}, \mathbf{v}) := \frac{1}{2m} \sum_{i=1}^m \left(\text{sign}(|\mathbf{a}_i^\top \mathbf{u}| - \tau) - \text{sign}(|\mathbf{a}_i^\top \mathbf{v}| - \tau) \right) \text{sign}(\mathbf{a}_i^\top \mathbf{u}) \mathbf{a}_i$$

Introduce the index sets

$$\mathcal{R}_{\mathbf{u}, \mathbf{v}} = \left\{ i \in [m] : \text{sign}(|\mathbf{a}_i^\top \mathbf{u}| - \tau) \neq \text{sign}(|\mathbf{a}_i^\top \mathbf{v}| - \tau) \right\},$$

$$\mathcal{L}_{\mathbf{u}, \mathbf{v}} = \left\{ i \in [m] : \text{sign}(\mathbf{a}_i^\top \mathbf{u}) \neq \text{sign}(\mathbf{a}_i^\top \mathbf{v}) \right\}.$$

$$\mathbf{h}(\mathbf{u}, \mathbf{v}) = \underbrace{\frac{1}{m} \sum_{i \in \mathcal{R}_{\mathbf{u}, \mathbf{v}}} \text{sign}(\mathbf{a}_i^\top (\mathbf{u} - \mathbf{v})) \mathbf{a}_i}_{:= \mathbf{h}_1(\mathbf{u}, \mathbf{v}) \text{ (main term)}} + \underbrace{\frac{1}{m} \sum_{i \in \mathcal{R}_{\mathbf{u}, \mathbf{v}} \cap \mathcal{L}_{\mathbf{u}, \mathbf{v}}} \left[\text{sign}(\mathbf{a}_i^\top (\mathbf{u} + \mathbf{v})) - \text{sign}(\mathbf{a}_i^\top (\mathbf{u} - \mathbf{v})) \right] \mathbf{a}_i}_{:= \mathbf{h}_2(\mathbf{u}, \mathbf{v}) \text{ (higher-order term)}}.$$

$$\mathbb{P}(i \in \mathcal{R}_{\mathbf{u}, \mathbf{v}} \cap \mathcal{L}_{\mathbf{u}, \mathbf{v}}) \ll \mathbb{P}(i \in \mathcal{R}_{\mathbf{u}, \mathbf{v}}) \implies |\mathcal{R}_{\mathbf{u}, \mathbf{v}} \cap \mathcal{L}_{\mathbf{u}, \mathbf{v}}| \ll |\mathcal{R}_{\mathbf{u}, \mathbf{v}}|.$$

Hence $\mathbf{h}_2$ is supported on a much smaller set, so it is a higher-order term.

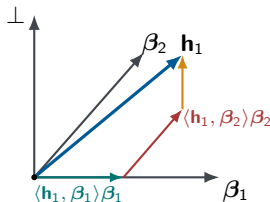
Orthogonal Decomposition

We decompose $\mathbf{h}_1(\mathbf{u}, \mathbf{v})$ along

$$\beta_1 := \frac{\mathbf{u} - \mathbf{v}}{\|\mathbf{u} - \mathbf{v}\|_2}, \beta_2 \in \text{span}(\mathbf{u}, \mathbf{v}), \beta_2 \perp \beta_1,$$

and the orthogonal directions $\text{span}(\mathbf{u}, \mathbf{v})^\perp$:

$$\mathbf{h}_1(\mathbf{u}, \mathbf{v}) = \langle \mathbf{h}_1(\mathbf{u}, \mathbf{v}), \beta_1 \rangle \beta_1 + \langle \mathbf{h}_1(\mathbf{u}, \mathbf{v}), \beta_2 \rangle \beta_2 + \mathbf{h}_1^\perp(\mathbf{u}, \mathbf{v}).$$



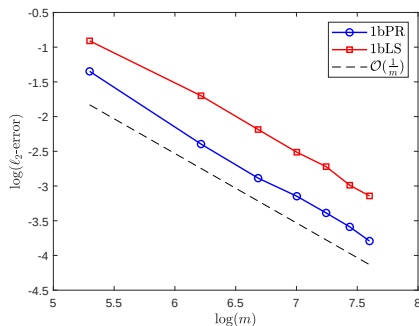
Then

$$\|\mathbf{u} - \mathbf{v} - \eta \mathbf{h}_1(\mathbf{u}, \mathbf{v})\|_2 \leq \underbrace{\left| \|\mathbf{u} - \mathbf{v}\|_2 - \eta \langle \mathbf{h}_1(\mathbf{u}, \mathbf{v}), \beta_1 \rangle \right|}_{:= \text{contraction}} + \underbrace{\left| \langle \mathbf{h}_1(\mathbf{u}, \mathbf{v}), \beta_2 \rangle \right| + \|\mathbf{h}_1^\perp(\mathbf{u}, \mathbf{v})\|_2}_{:= \text{fluctuation}}.$$

$$\begin{aligned} \left| \|\mathbf{u} - \mathbf{v}\|_2 - \eta \langle \mathbf{h}_1(\mathbf{u}, \mathbf{v}), \beta_1 \rangle \right| &= \|\mathbf{u} - \mathbf{v}\|_2 \left| 1 - \frac{\eta}{\|\mathbf{u} - \mathbf{v}\|_2} \frac{1}{m} \sum_{i=1}^m |\mathbf{a}_i^\top \beta_1| \mathbf{1}(i \in \mathcal{R}_{\mathbf{u}, \mathbf{v}}) \right| \\ &\leq \|\mathbf{u} - \mathbf{v}\|_2 \left[\underbrace{\frac{\eta}{\|\mathbf{u} - \mathbf{v}\|_2} \left| \frac{1}{m} \sum_{i=1}^m \{ |\mathbf{a}_i^\top \beta_1| \mathbf{1}(i \in \mathcal{R}_{\mathbf{u}, \mathbf{v}}) - \mathbb{E}[|\mathbf{a}_i^\top \beta_1| \mathbf{1}(i \in \mathcal{R}_{\mathbf{u}, \mathbf{v}})] \} \right|}_{\text{Bernstein's inequality - } O(\|\mathbf{u} - \mathbf{v}\|_2) \text{ variance}} + \underbrace{\left| 1 - \frac{\eta \mathbb{E}[|\mathbf{a}_i^\top \beta_1| \mathbf{1}(i \in \mathcal{R}_{\mathbf{u}, \mathbf{v}})]}{\|\mathbf{u} - \mathbf{v}\|_2} \right|}_{\text{integral calculation - decide } \eta} \right] \end{aligned}$$

Experiments

Synthetic data: $O(m^{-1})$ decay rate



blue curve: recover $\mathbf{x} \in S^{29}$ from $\mathbf{y} = \text{sign}(|\mathbf{Ax}| - 1)$

red curve: recover $\mathbf{x} \in S^{29}$ from $\mathbf{y} = \text{sign}(\mathbf{Ax})$

Natural image (Coded Diffraction Patterns)



(a) Original image: Milky Way Galaxy.



(b) Recovered image after spectral initialization: relative error = 0.270, PSNR = 25.14.



(c) Recovered image after gradient descent : relative error = 0.029, PSNR = 44.65.

Figure: Recovering the $1080 \times 1980 \times 3$ Milky Way Galaxy image from phaseless bits produced by CDP with $L = 64$ random patterns.

Sparse Case

One-Bit Sparse Phase Retrieval (1bSPR)

Recover a sparse signal $\mathbf{x} \in \Sigma_k^n \cap \mathbb{A}_{\alpha,\beta}$ from one-bit phaseless measurements $\mathbf{y} = \text{sign}(|\mathbf{A}\mathbf{x}| - \tau)$.

Statistical Optimal Rate

$$\inf_{\hat{\mathbf{x}}} \sup_{\mathbf{x} \in \Sigma_k^n \cap \mathbb{A}_{\alpha,\beta}} \text{dist}(\hat{\mathbf{x}}, \mathbf{x}) = \tilde{\Theta}\left(\frac{k}{m}\right).$$

Phases are inessential in one-bit compressed sensing.

Optimal Algorithm - Iterative Hard Thresholding

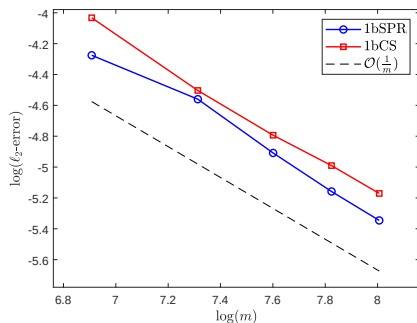
$\mathbf{x}_{t+1} = T_k(\mathbf{x}_t - \eta \mathbf{h}(\mathbf{x}_t, \mathbf{x}))$ locally achieves the optimal error under $m = \tilde{\Omega}(k)$

Restricted Approximate Invertibility Condition (RAIC) (Abdalla et al., 2026; Chen et al., 2025)

$$\|T_{2k}(\mathbf{u} - \mathbf{v} - \eta \mathbf{h}(\mathbf{u}, \mathbf{v}))\|_2 \leq c_1 \|\mathbf{u} - \mathbf{v}\|_2 + \sqrt{\tilde{O}\left(\frac{k}{m}\right)} \|\mathbf{u} - \mathbf{v}\|_2 + \tilde{O}\left(\frac{k}{m}\right),$$

$$\forall \mathbf{u}, \mathbf{v} \in \Sigma_k^n \cap \mathbb{A}_{\alpha,\beta}, \quad \|\mathbf{u} - \mathbf{v}\|_2 \lesssim \frac{1}{\sqrt{\log(m/k)}}.$$

Synthetic data: $O(m^{-1})$ decay rate



blue curve (1bSPR): recover $\mathbf{x} \in \Sigma_3^{500,*}$ from $\mathbf{y} = \text{sign}(|\mathbf{Ax}| - 1)$

red curve (1bCS): recover $\mathbf{x} \in \Sigma_3^{500,*}$ from $\mathbf{y} = \text{sign}(\mathbf{Ax})$

Conclusion

Conclusion

1bCS	1bPR	1bSPR
$\tilde{\Theta}\left(\frac{k}{m}\right)$ Jacques et al. (2013)	$\tilde{\Theta}\left(\frac{n}{m}\right)$ our work	$\tilde{\Theta}\left(\frac{k}{m}\right)$ our work
NBIHT $m = \tilde{\Omega}(k)$ Matsumoto and Mazumdar (2024)	ReLU GD $m = \tilde{\Omega}(n)$ our work	ReLU IHT $m = \tilde{\Omega}(k^2)$ our work

Phase is inessential for 1bCS

“One-Bit Phase Retrieval: Optimal Rates and Efficient Algorithms”

IEEE Transactions on Information Theory, 2026

<https://arxiv.org/abs/2405.04733>

Thank You!

Questions?

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